

The Bishop-Phelps-Bollobas Moduli of $L(H)$ Are Maximal

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Source Question

The source paper is Chica–Kadets–Martin–Moreno–Pulido–Rambla–Barreno, *Bishop-Phelps-Bollobas moduli of a Banach space*, arXiv:1304.0376. It proves maximal Bishop-Phelps-Bollobas moduli for many spaces, including unital complex C^* -algebras with nontrivial center, and then asks whether the same is true for $L(H)$. The paper notes that its centralizer and M-ideal criteria do not apply to $L(H)$.

BISHOP-PHELPS-BOLLOBÁS MODULI OF A BANACH SPACE

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It would be interesting to see whether the algebra $L(H)$ for a finite- or infinite-dimensional Hilbert space H has the maximum Bishop-Phelps-Bollobás moduli. None of the results of this section applies to it since its center is trivial and, despite it contains $K(H)$ as an M -ideal, there is no element $x_0 \in L(H)$ satisfying the requirements of Corollary 4.9 (see [?, page 538]). Let us also comment that the bidual of $L(H)$ is a C^* -algebra with non-trivial centralizer, so $\Phi_{L(H)^{**}}(\delta) = \Phi_{L(H)^{**}}^S(\delta) = \sqrt{2\delta}$ for every $\delta \in (0, 1/2]$. If there is $\delta \in (0, 1/2]$ such that $\Phi_{L(H)}(\delta) < \sqrt{2\delta}$, then this would be an example when the inequality in Proposition 3.6 is strict.

We finish this section with two pictures: one with the Bishop-Phelps-Bollobás moduli of $\mathbb{R}$, $\mathbb{C}$ and $\ell_\infty^{(2)}$, and another one with the corresponding values of the spherical Bishop-Phelps-Bollobás moduli.



This packet gives a positive answer for every complex Hilbert space H with $\dim H \geq 2$.

Definitions

For a Banach space X , write

$$\Pi(X) = \{(y, y^*) \in S_X \times S_{X^*} : y^*(y) = 1\}.$$

The Bishop-Phelps-Bollobas modulus $\Phi_X(\delta)$ is the least radius which approximates every pair $(x, x^*) \in B_X \times B_{X^*}$ satisfying $\operatorname{Re} x^*(x) > 1 - \delta$ by some element of $\Pi(X)$. The spherical modulus $\Phi_X^S(\delta)$ is defined in the same way with $(x, x^*) \in S_X \times S_{X^*}$.

The source paper proves the universal upper bound

$$\Phi_X^S(\delta) \leq \Phi_X(\delta) \leq \sqrt{2\delta} \quad (0 < \delta < 2).$$

Therefore, to prove maximality it suffices to prove the matching lower bound for Φ_X^S .

For unit vectors $\xi, \eta \in H$, write

$$\omega_{\eta, \xi}(a) = \langle a\xi, \eta \rangle \quad (a \in B(H)).$$

When $\eta = \xi$, write $\omega_\xi = \omega_{\xi, \xi}$.

Theorem

Theorem 1. *Let H be a complex Hilbert space with $\dim H \geq 2$, and let $X = L(H) = B(H)$ with the operator norm. Then*

$$\Phi_X(\delta) = \Phi_X^S(\delta) = \sqrt{2\delta} \quad (0 < \delta \leq 1/2).$$

Proof

We use two elementary Hilbert-operator facts. For $T \in B(H)$, let $a_2(T)$ denote the second approximation number,

$$a_2(T) = \inf\{\|T - F\| : \text{rank } F < 2\}.$$

In finite dimension this is the second singular value.

Lemma 1. *Let $0 < \varepsilon \leq 1$, let p be a rank-one orthogonal projection on H , and put*

$$x = (1 - \varepsilon)I + \varepsilon p.$$

If $y \in B(H)$, $\|y\| = 1$, and $\|x - y\| < \varepsilon$, then

$$a_2(y) < 1.$$

*Consequently, 1 is a simple top singular value of y : the spectral projection of $|y| = (y^*y)^{1/2}$ corresponding to the eigenvalue 1 is rank one.*

Proof. Since $x - p = (1 - \varepsilon)(I - p)$, we have

$$a_2(x) \leq \|x - p\| = 1 - \varepsilon.$$

The approximation numbers are Lipschitz in the operator norm, hence

$$a_2(y) \leq a_2(x) + \|x - y\| < 1 - \varepsilon + \varepsilon = 1.$$

As $\|y\| = 1$, the strict inequality $a_2(y) < 1$ puts the norm of y above the second approximation number. By the Hilbert-space min-max characterization of approximation numbers, the singular values of y above $a_2(y)$ form a one-dimensional top singular space. Hence the singular value 1 is attained and has multiplicity one. $\square$

Lemma 2. *Let $y \in B(H)$ satisfy $\|y\| = 1$ and $a_2(y) < 1$. If $\varphi \in B(H)^*$, $\|\varphi\| = 1$, and $\varphi(y) = 1$, then φ is a rank-one vector functional:*

$$\varphi(a) = \omega_{\eta, \xi}(a) = \langle a\xi, \eta \rangle \quad (a \in B(H))$$

for unit vectors $\xi, \eta \in H$.

Proof. By Lemma 1's conclusion applied to y , the spectral projection e of $|y|$ for the eigenvalue 1 has rank one. Let $e = \xi \otimes \xi$, $\|\xi\| = 1$. Since $a_2(y) < 1$, there is $0 \leq \beta < 1$ such that

$$y^*y \leq e + \beta^2(I - e).$$

Let $|\varphi|$ be the right absolute value of φ in the polar decomposition of functionals on the C^* -algebra $B(H)$, so that the C^* -Cauchy-Schwarz inequality gives

$$1 = |\varphi(y)|^2 \leq |\varphi|(y^*y) \leq 1.$$

Thus $|\varphi|(y^*y) = 1$. From the displayed bound on y^*y ,

$$1 = |\varphi|(y^*y) \leq |\varphi|(e) + \beta^2|\varphi|(I - e) = \beta^2 + (1 - \beta^2)|\varphi|(e).$$

Since $\beta < 1$, this forces $|\varphi|(e) = 1$, hence $|\varphi|(I - e) = 0$. By Cauchy-Schwarz, $|\varphi|$ vanishes on the off-corner terms $(I - e)B(H)$ and $B(H)(I - e)$. Therefore

$$|\varphi|(a) = |\varphi|(eae) = \langle a\xi, \xi \rangle \quad (a \in B(H)).$$

Again by C^* -Cauchy-Schwarz,

$$|\varphi(a)|^2 \leq |\varphi|(a^*a) = \|a\xi\|^2.$$

So $\varphi(a)$ depends only on the vector $a\xi$, and defines a norm-one functional on H . Hence there is a unit vector $\eta \in H$ such that

$$\varphi(a) = \langle a\xi, \eta \rangle \quad (a \in B(H)).$$

The equality $\varphi(y) = 1$ then gives $\langle y\xi, \eta \rangle = 1$, so necessarily $\eta = y\xi$. In particular, φ is rank one. $\square$

Lemma 3. Let $0 < \varepsilon \leq 1$. Choose orthonormal vectors $\xi, \zeta \in H$, let $p = \xi \otimes \xi$, and put

$$x = (1 - \varepsilon)I + \varepsilon p, \quad \rho = (1 - \varepsilon/2)\omega_\xi + (\varepsilon/2)\omega_\zeta.$$

If $(y, \varphi) \in \Pi(B(H))$, then

$$\max\{\|x - y\|, \|\rho - \varphi\|\} \geq \varepsilon.$$

Proof. Assume $\|x - y\| < \varepsilon$. By Lemma 1, $a_2(y) < 1$. By Lemma 2, the norming functional φ is a rank-one vector functional.

The functional ρ is normal and is represented by the positive trace-class operator

$$(1 - \varepsilon/2)\xi \otimes \xi + (\varepsilon/2)\zeta \otimes \zeta,$$

whose singular values are $1 - \varepsilon/2, \varepsilon/2, 0, \dots$. A rank-one norm-one vector functional is represented by a trace-class operator with singular values $1, 0, 0, \dots$. Since both functionals are normal, their distance as elements of $B(H)^*$ is the trace norm of the difference. The Ky Fan–Mirsky singular-value inequality therefore gives

$$\|\rho - \varphi\| \geq |(1 - \varepsilon/2) - 1| + |\varepsilon/2 - 0| = \varepsilon.$$

Thus whenever $\|x - y\| < \varepsilon$, one has $\|\rho - \varphi\| \geq \varepsilon$. The lemma follows. $\square$

We now prove the theorem. Fix $0 < \delta \leq 1/2$, choose $0 < \gamma < \delta$, and set $\varepsilon = \sqrt{2\gamma}$. Then $0 < \varepsilon \leq 1$. With x and ρ as in Lemma 3, we have

$$\|x\| = 1, \quad \|\rho\| = 1,$$

and, since $\zeta \perp \xi$,

$$\rho(x) = (1 - \varepsilon/2) \cdot 1 + (\varepsilon/2)(1 - \varepsilon) = 1 - \varepsilon^2/2 = 1 - \gamma > 1 - \delta.$$

Thus $(x, \rho) \in S_X \times S_{X^*}$ is an admissible spherical test pair for $\Phi_X^S(\delta)$. Lemma 3 says that its d_∞ -distance from $\Pi(X)$ is at least $\varepsilon = \sqrt{2\gamma}$. Hence

$$\Phi_X^S(\delta) \geq \sqrt{2\gamma}.$$

Letting $\gamma \uparrow \delta$, we get

$$\Phi_X^S(\delta) \geq \sqrt{2\delta}.$$

The universal upper bound gives

$$\Phi_X^S(\delta) \leq \Phi_X(\delta) \leq \sqrt{2\delta}.$$

Therefore both moduli equal $\sqrt{2\delta}$.

What Changed From the M_2 Packet

The earlier $M_2(\mathbb{C})$ packet used $\text{diag}(1 - \varepsilon, 1)$. In higher dimension, the direct analogue $\text{diag}(1 - \varepsilon, 1, \dots, 1)$ has a large top singular face, so norming functionals can spread over that face.

The present proof uses the reversed witness

$$(1 - \varepsilon)I + \varepsilon p,$$

whose top singular value is simple and separated from all other singular values by exactly ε . This forces every sufficiently close norm-attaining contraction to have a rank-one norming functional, reducing the problem to the same trace-norm distance calculation as in the M_2 case.

Human Review Recommendation

The proof depends on two standard facts that are worth checking carefully:

1. If $a_2(y) < \|y\| = 1$, then 1 is a simple top singular value of y .
2. A norm-one functional $\varphi \in B(H)^*$ with $\varphi(y) = 1$ must be a vector functional when the top singular value of y is simple.

Both facts are proved above, but they are the right places for a reviewer to look first.

References

- [1] M. Chica, V. Kadets, M. Martin, S. Moreno-Pulido and F. Rambla-Barreno, *Bishop-Phelps-Bollobas moduli of a Banach space*, J. Math. Anal. Appl. 412 (2014), 697–719; arXiv:1304.0376.
- [2] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics 169, Springer, 1997.